PROJ 3002 Special Project II
Baganetic

TOUR GUIDE ASSISTANT FOR BAGAN PAGODAS



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PROJECT MEMBERS AND ROLES

1. Khaing Thazin Tun	(2019-MIIT-CSE-014)	Team Leader & Backed Developer
2. Kyaw Zay Ya Hein	(2022-MIIT-CSE-014)	Fronted Developer & Database
3. Htun Lei Lei Win	(2022-MIIT-CSE-033)	Data Researcher & Documentation
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2 Abstract of the System

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INTRODUCTION



- Web-based platform for preserving and promoting Bagan’s heritage
- User-friendly system with temple profiles, maps, and cultural details
- Enhances accessibility and cultural education through digital heritage

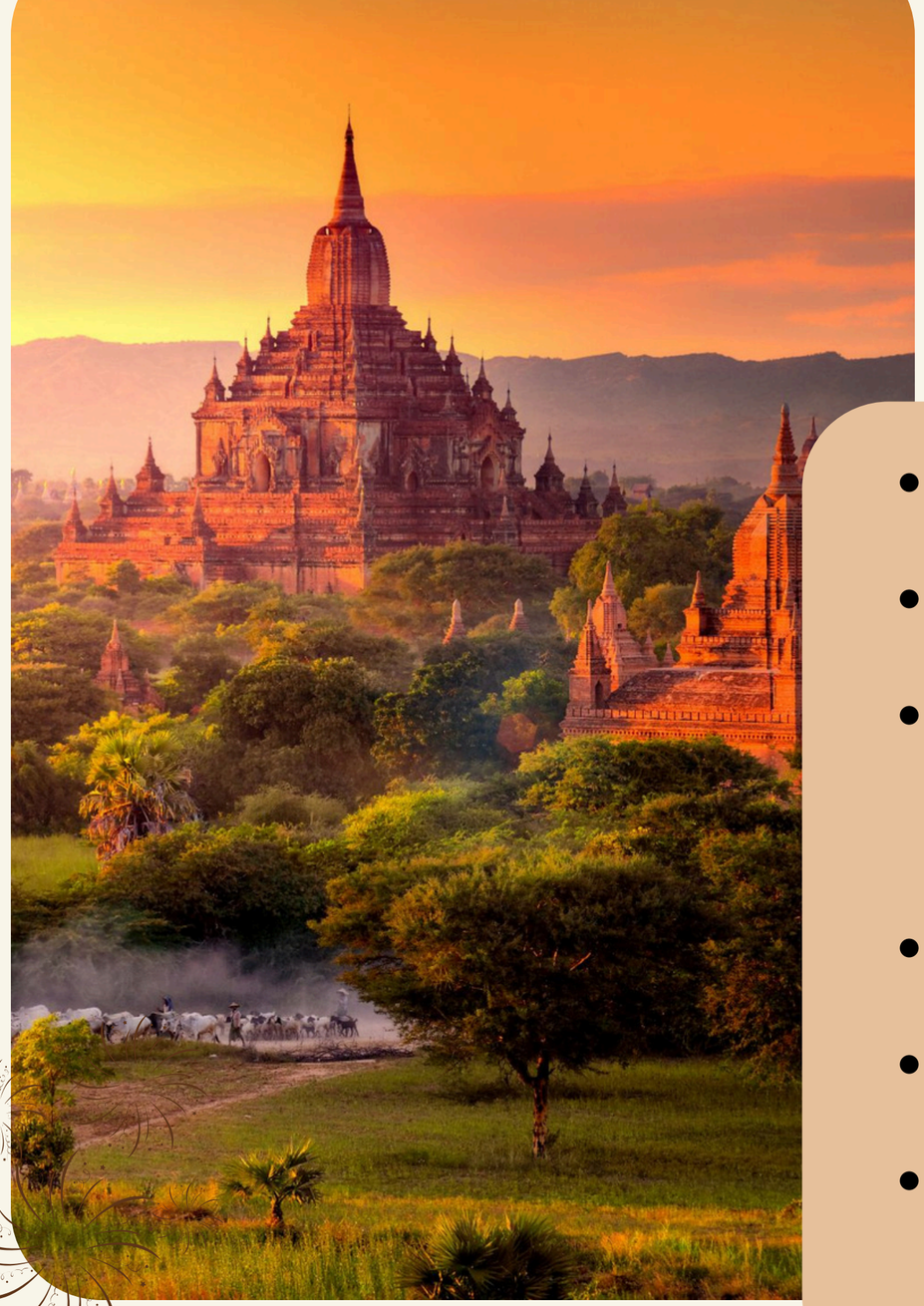
PROBLEM STATEMENT

(WHY OUR PROJECT IS UNIQUE?)



Bagan has over 2,000 pagodas, making it hard for the tourists to find the best route.

Our system solves this problem with an innovative approach called A* Algorithm.



ABSTRACT OF THE SYSTEM

- Web app to explore and preserve Bagan’s temples
- Interactive map with pagoda info
- Built with HTML, CSS, JavaScript, Python (Flask), and MongoDB
- A* pathfinding for optimized routes
- AI chatbot for instant travel tips
- Enhances tourist experience and promotes Myanmar’s heritage

PURPOSE OF DEVELOPING BAGANETIC



- To implement an easy and enjoyable system for users
- To provide an easy-to-use knowledge hub of temples
- To help tourists easily navigate between landmarks in Bagan
- To save time, enhance user travel experience in cultural heritage zones

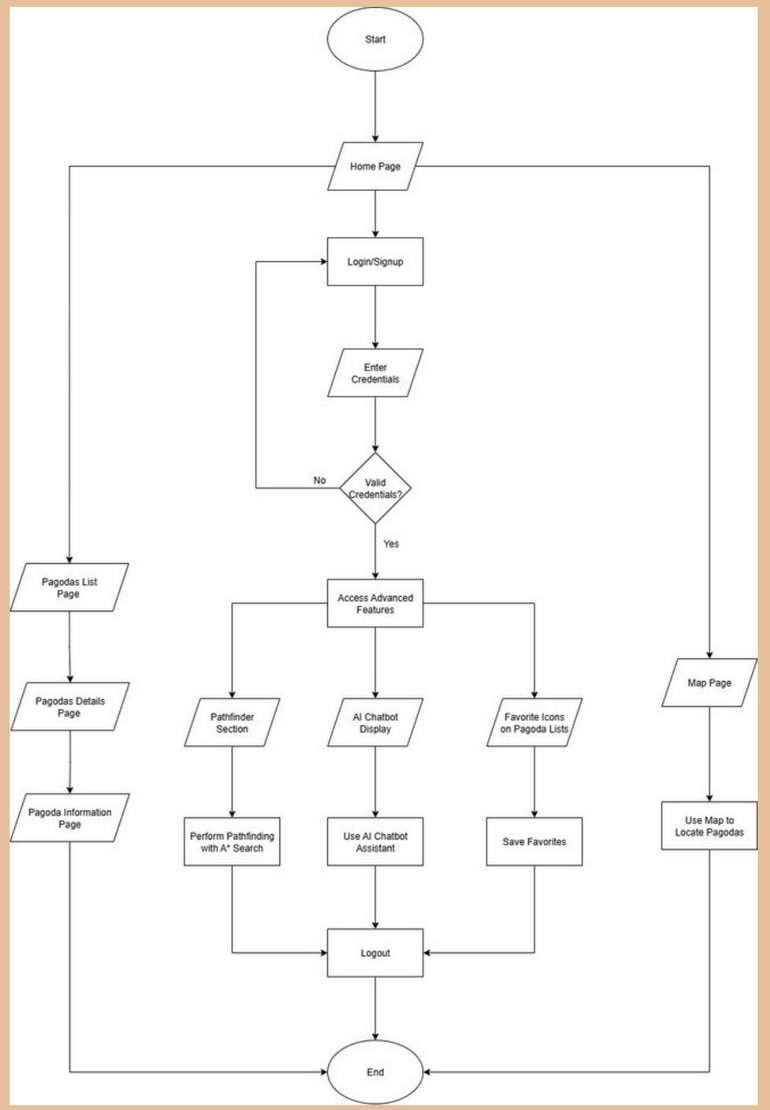
USER SIDE FEATURES

- 🏯 Explore Temple Catalog
- 🔍 Advanced Search & Filter
- 🗺️ Interactive Map Navigation
- 🧭 Shortest Path Finder
- 🤖 AI Chatbot Assistant
- ⭐ Favorite & Visited Marking System
- 👤 User Profile Management
- 🌏 Bilingual Support (English/Myanmar)

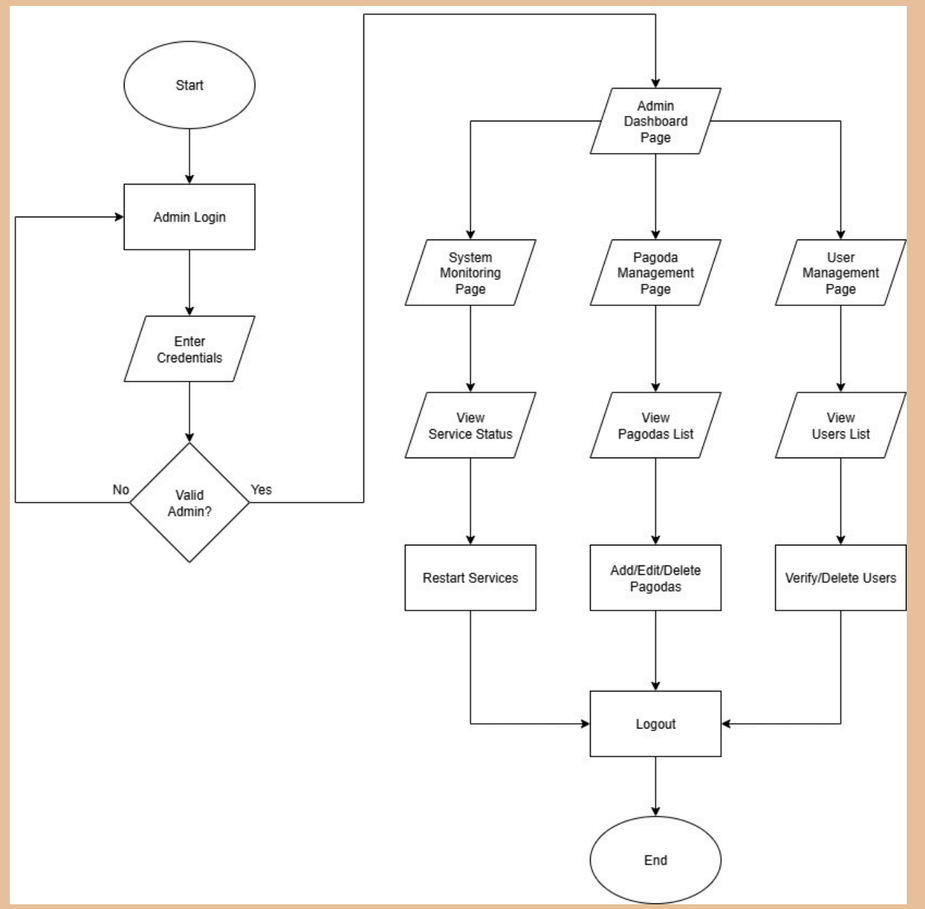
ADMIN SIDE FEATURES

- 🔐 Secure Admin Login
- 📊 Dashboard Overview
- 🏗️ Pagoda Management
- 👥 User Management
- ⚙️ System Health Monitoring
- 📜 Admin Activity Log

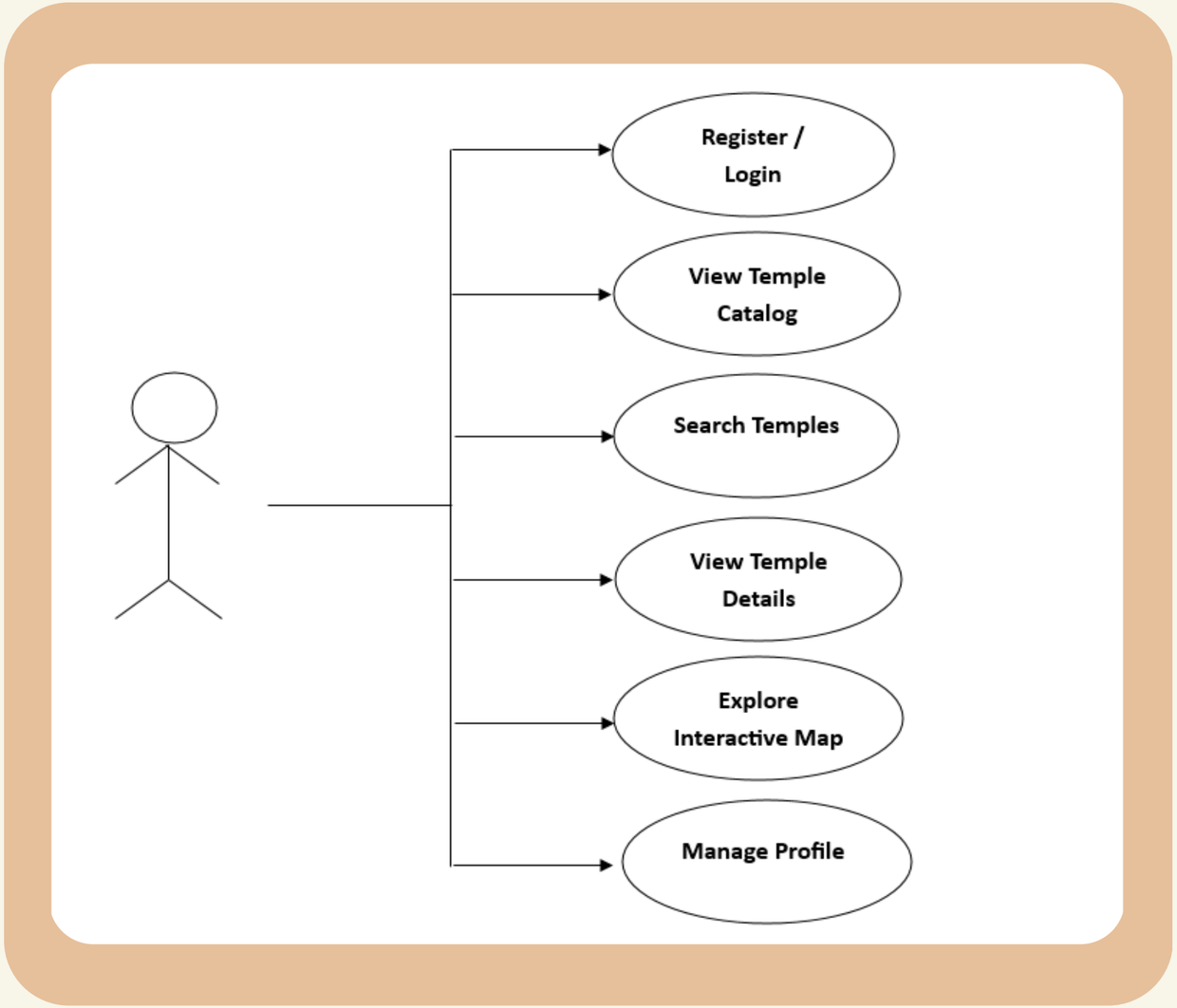
FLOWCHART FOR USER



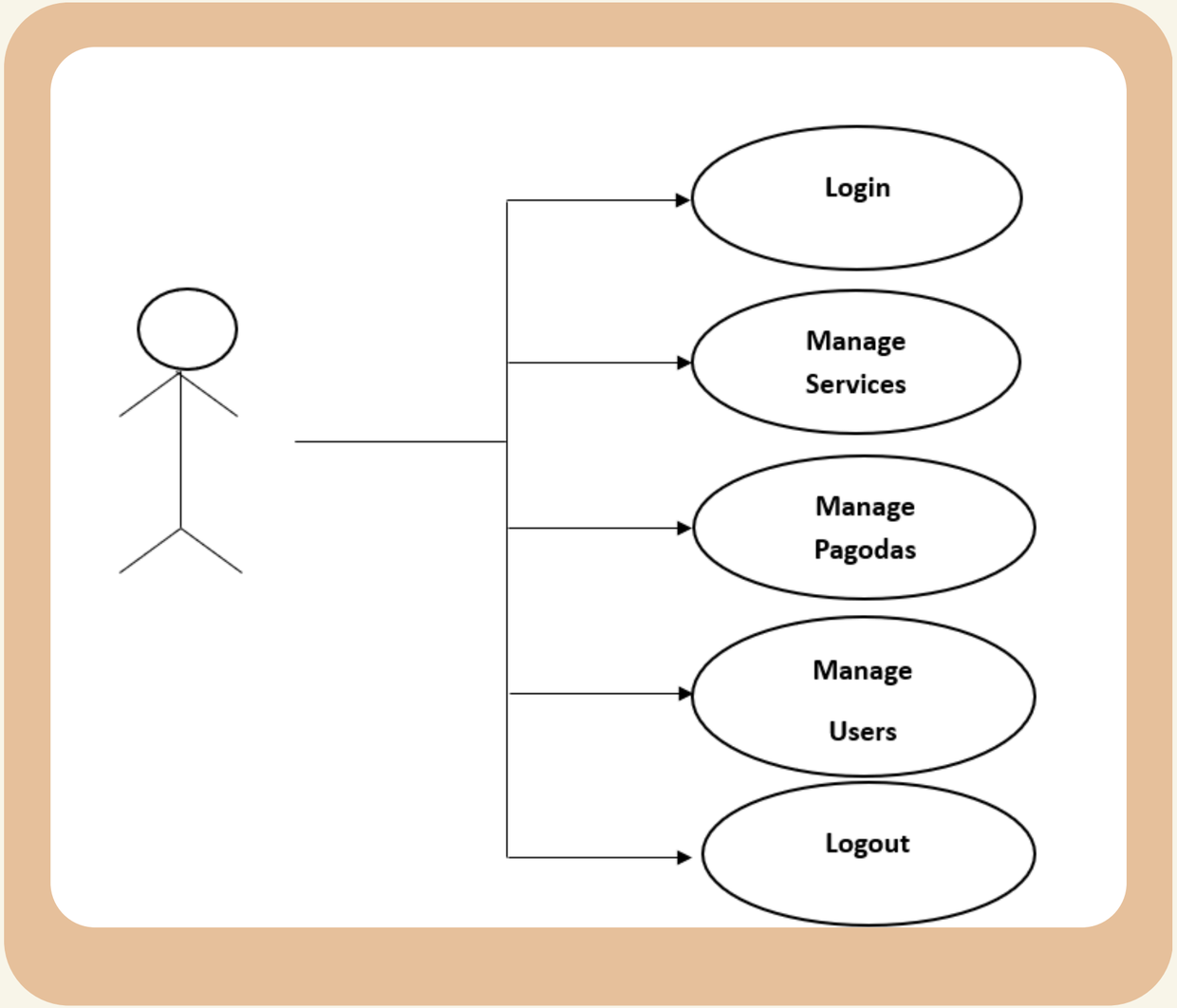
FLOWCHART FOR ADMIN



USE CASE DIAGRAM FOR USER



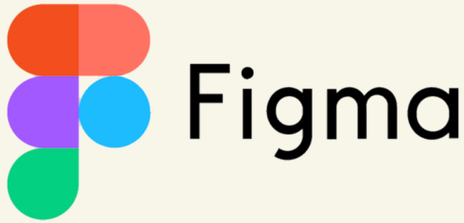
USE CASE DIAGRAM FOR ADMIN



TECHNOLOGY



FRONTENED



UI /UX DESIGN

DATABASE



BACKENED



WHAT IS A* ALGORITHM?

- A* is a search algorithm used in maps, GPS, and games.
- It finds the shortest path between two points.
- Combination of Dijkstra's algorithm and Greedy Best-First Search algorithm



WHY CHOOSE A*

A* is better than other algorithms:

vs Dijkstra's Algorithm: Dijkstra only use g(n) and explores everywhere equally; A* uses heuristic to go towards the goal.

vs Greedy Best-First Search: Greedy only uses h(n) (can miss optimal path); A* balances g(n) and h(n).

COMPARISON OF ALGORITHMS

Algorithm	Pros	Cons
Dijkstra’s	Always finds shortest path	Explores too many nodes → slow
Greedy Best-First	Very fast, simple	May fail to find shortest path/ risk to take a path that does not bring to the goal
A*	Shortest path + efficient	Need good heuristic, use more memory

THE A* FORMULA

$f(n) = g(n) + h(n)$

- g(n): The cost to reach the current point from the start.
- h(n): The heuristic (estimated distance) from the current point to the goal.
- f(n): The total estimated cost.

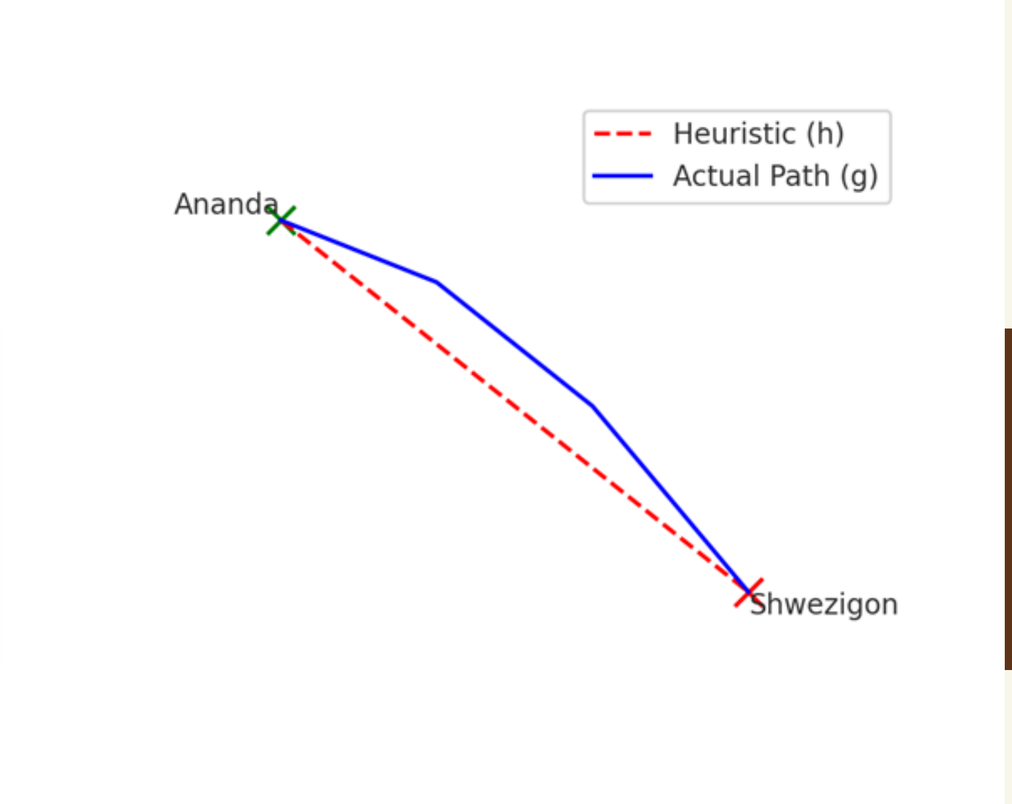
HEURISTIC (H(N))

- In our project, we use GPS coordinates of pagodas.
- To estimate distance, we use the Haversine formula (straight-line distance).
- Example:
  - Ananda → Thambula
  - h(n) ≈ 2.38 miles (straight line)

This helps the algorithm move in the right direction faster.

HEURISTIC EXAMPLE

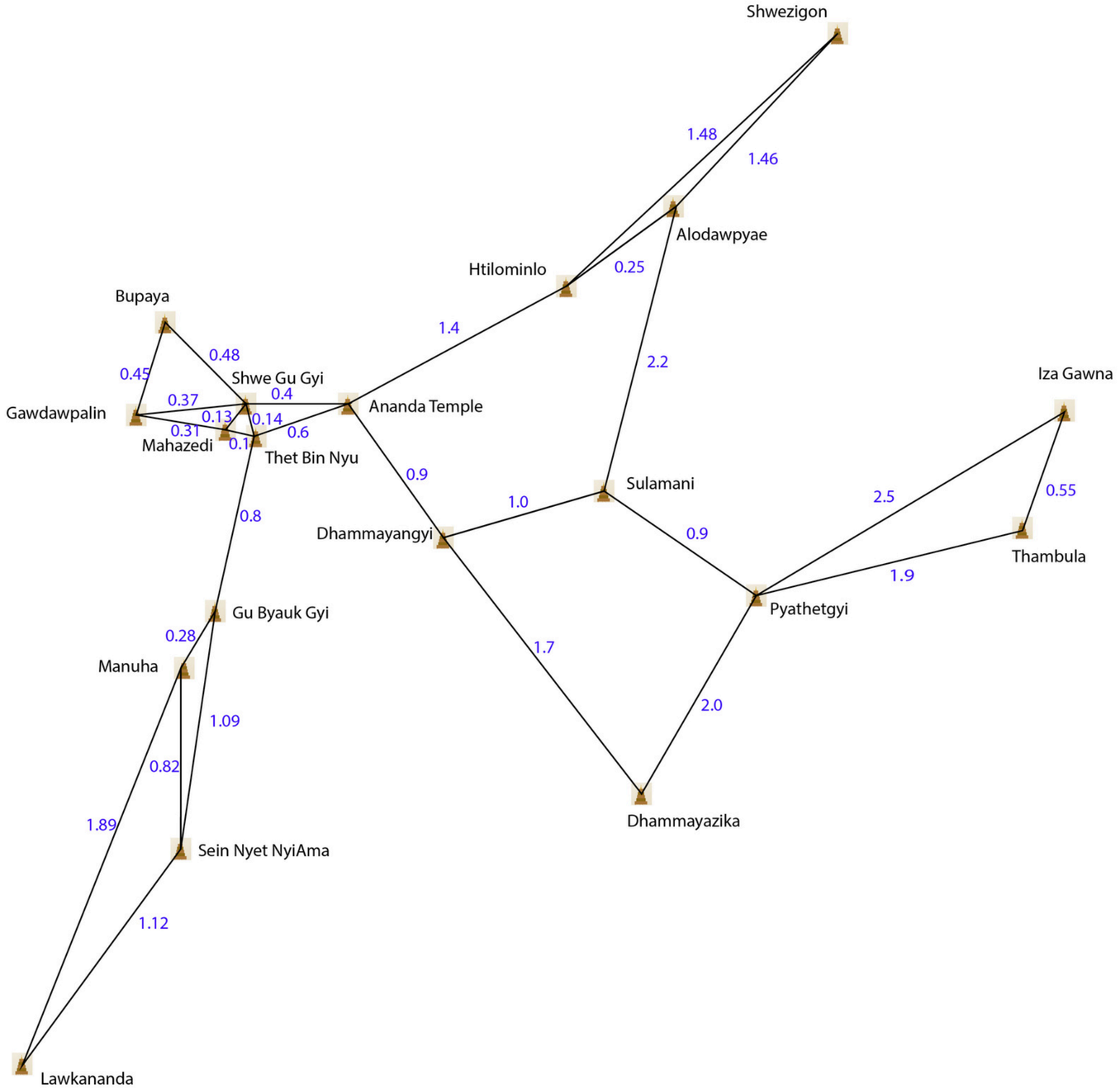
- Straight red line = heuristic (h)
- Blue curved line = real road (g)
- Heuristic guides us in the right direction faster



MAP OF BAGAN


N
W
E
S
Ayeyarwady River
Shwe Zi Gone Pagoda
Kyan Sit Thar Village
Tansaung Kyaung
Wet Kyi Inn Village
Nyaung Oo Rd.
Gubyaukkyi
Nyang U Town
Gubyauknge
Anawrahta Rd.
Nyaung Oo - Kyaukadaung Rd.
Upali Thein
Htilominlo
Bu Phaya
Tharaba Gateway
Mahabodi Pagoda
Alotawpyi
Oak Kyaung Gyi
Ananda
Gawdawpalin
Bagan Golf Course
Old Bagan
To Mount Popa
To Airport
Shwegugyi
Thabyinyu
Mingalar Kyaung
Nan Myint Tower
Nanda Pyin Nya
Thambula
Nathlaung Kyaung
Sulamani
Tharaba Gateway
Mya Zedi
Shwe San Daw
Payathonzu
Thamanpaya
Mingalar Zedi
Dhamayangyi
Minnathu Village
Hpyauk Seik Pin Village
Myat Lay Rd.
Manuha
Nanpaya
Thamudi
West Pwa Saw Village
Apeyadana
Nagayon
Tamani Pagoda
Hsu Taung Pyi
East Pwa Saw Village
Seinnyet Nyima & Ama
Tamani Oak Kyaung
Dhammayazaka Pagoda
Kodangyi Village
Thingaraza Pagoda
New Bagan
Thiri Malar St.
Hninsk St.
Paneayoe St.
Khawamyo St.
Khattar St.
Thuhtakan Village
Nweni St.
Khayoe St.
Sweltaw St.
Gangaw St.
Yuzana St.
Sein Pan St.
Gandamar St.
Thaphaye St.
Lawkananda Pagoda
Thazin St.
Thawka St.

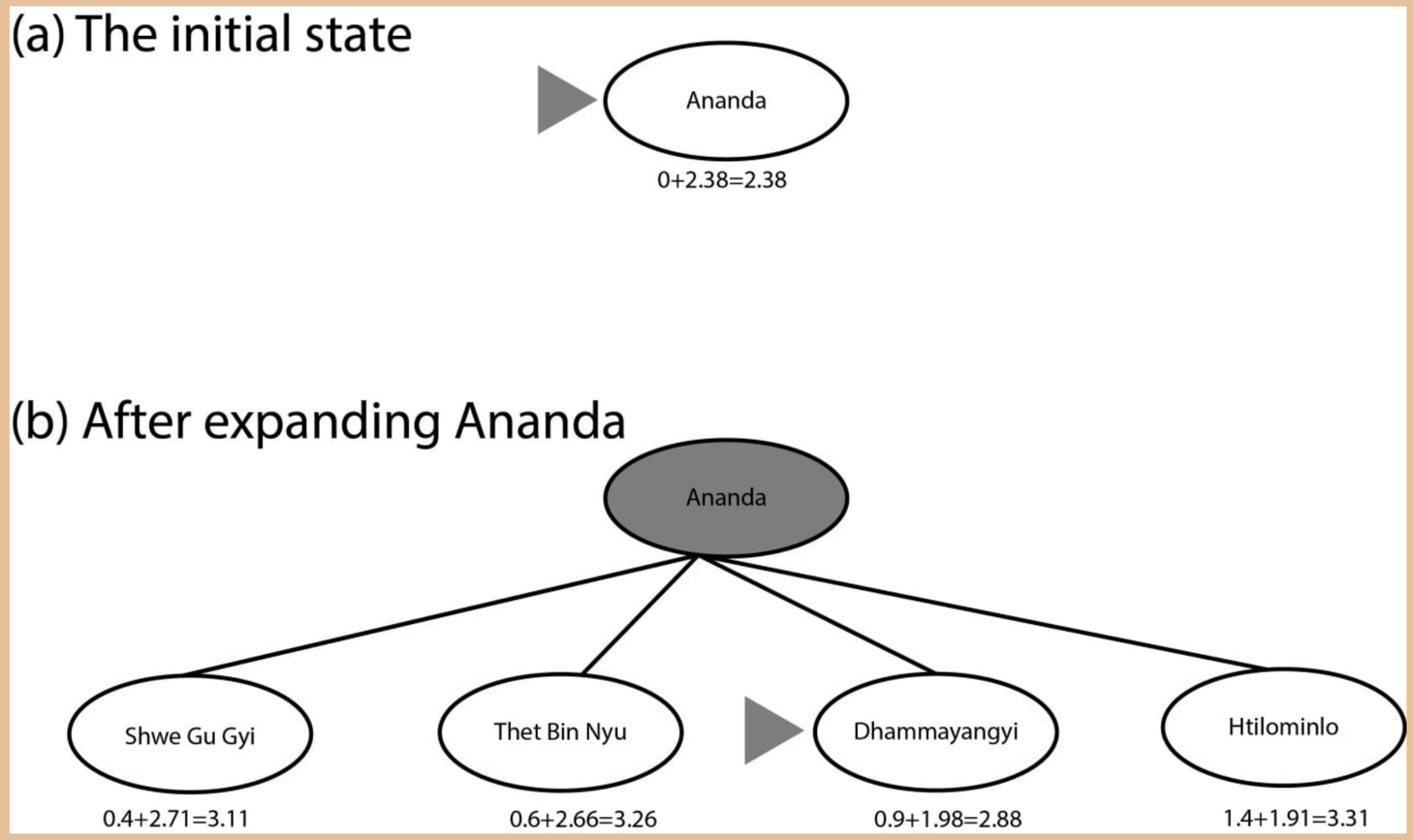
Simplified Road Map of Bagan Pagodas(with distances in miles)



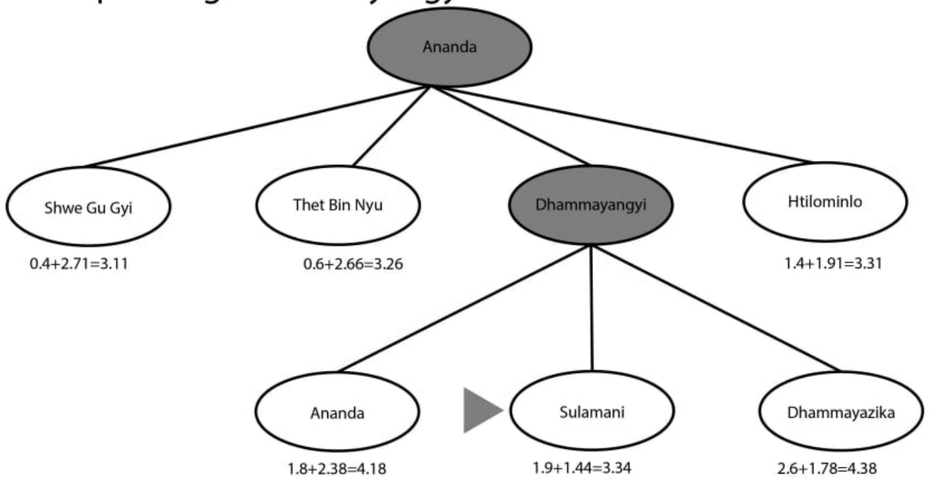
HEURISTIC DISTANCE TO THAMBULA

Pagoda	h(n) to Thambula
Ananda	2.38
Dhammayangyi	1.98
Sulamani	1.44
Dhammayazika	1.78
Pyathetgyi	0.96
shwe gu gyi	2.72
That Bin Nyu	2.66
Htilominlo	1.91
Alodawpyae	1.66
Iza Gawna	0.55
Bupaya	3.09
Gawdawpalin	3.07
Mahazedi	2.76
Gu Byauk Gyi	2.79
Manuha	2.94
Sein Nyet NyiAma	3.23
Lawkananda	4.21
Thambula	0.00
Shwezigon	2.34

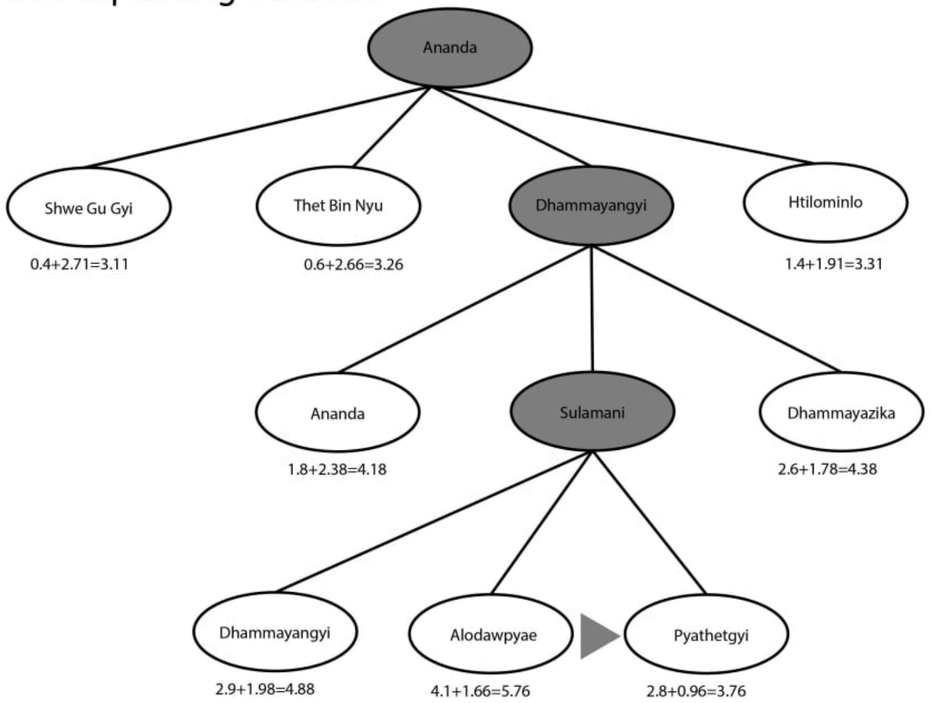
Stages in an A* search for Thambula (Nodes are labeled with g+h = f)



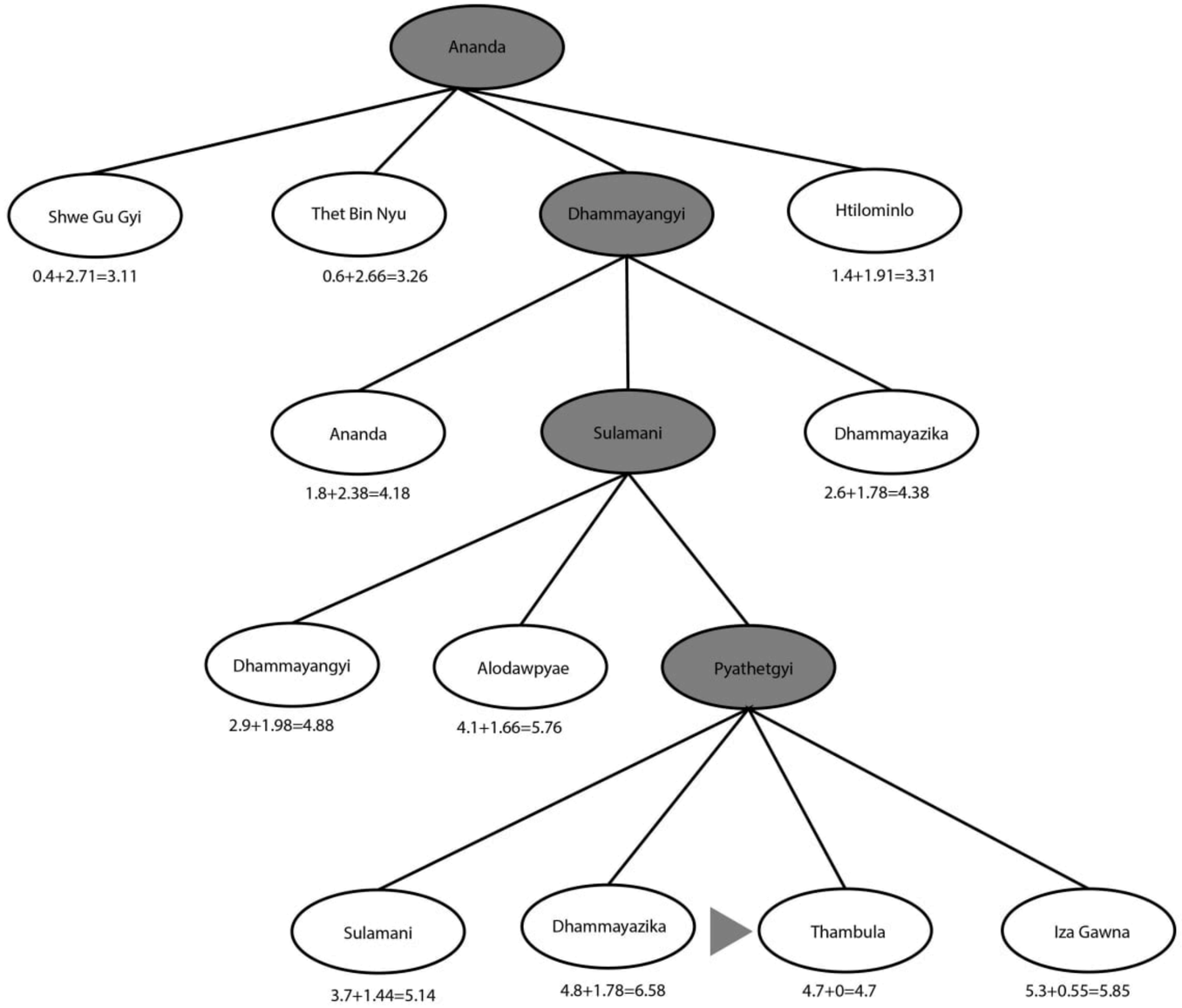
(c) After expanding Dhammayangyi



(d) After expanding Sulamani



(e) After expanding Pyathetgyi



Shortest path result:

Ananda → Dhammayangyi → Sulamani → Pyathetgyi → Thambula = 4.7 miles

IDENTIFICATION AND RESOLUTION OF BUGS AND ISSUES

Problem / Bug Identified

- MongoDB Connection Failures
- Missing Pagoda Data – Search sometimes returned blank because no matching pagoda was found.
- Cluttered Map Popups – CSS-only popups made info hard to read and not interactive.
- A* Algorithm Implementation Bugs
- Server Configuration Issues : Port Conflicts

Resolution / Improvement

- Fallback Data Loading from JavaScript files when MongoDB unavailable
- Error Handling – Displayed “No pagoda found” message instead of blank results.
- Switched to Leaflet Popups – Interactive, clean info windows on the map.
- Created improved_pathfinder.py with enhanced A* implementation
- Port Conflict Detection and resolution

AFTER MIDTERM IMPROVEMENTS

🗺️ Shortest Path Finder
🤖 AI Chatbot
⭐ Favorite Marking System
🛠️ Admin Dashboard

FUTURE ENHANCEMENT

- 🌐Mobile App Development
- 🤖AI-Powered Recommendations
- 🗣️Multi-Language Support
- ☁️Integration with External Services
- 📊User Feedback & Analytics

CONCLUSION

- Simplifies navigation with A* shortest path algorithm
- Enhances tourism experience with interactive maps & pagoda info
- Provides a reliable platform for future improvements
- Supports digital preservation of Myanmar’s heritage

REFERENCES

- Book : Artificial Intelligence - A Modern Approach (3rd Ed)
- Leaflet JS : https://leafletjs.com/reference.html
- MongoDB : https://www.mongodb.com/docs/manual/

THANK YOU

for your attention



LET'S GET INTO SEEING BAGENTIC IN ACTION

DISCUSSION SESSION